

EKS

Elastic kubernetes service

Go to eks

Create a cluster

Cluster configuration [Info](#)

Name
Enter a unique name for this cluster. This property cannot be changed after the cluster is created.

The cluster name should begin with letter or digit and can have any of the following characters: the set of Unicode letters, digits, hyphens and underscores. Maximum length of 100.

Cluster service role [Info](#)
Select the IAM role to allow the Kubernetes control plane to manage AWS resources on your behalf. This property cannot be changed after the cluster is created. To create a new role, follow the instructions in the [Amazon EKS User Guide](#).

[Create a role in IAM console](#)

Kubernetes version settings

(Create a role in iam console -> next -> next)

Kubernetes version settings

Kubernetes version [Info](#)
Select Kubernetes version for this cluster.

Upgrade policy [Info](#)
Choose one of the following options. You can switch the setting later while the standard support period is in effect.

☐ **Extended**
This option supports the Kubernetes version for 26 months after the release date. The extended support period has an additional hourly cost that begins after the standard support period ends. When extended support ends, your cluster will be auto upgraded to the next version.

☒ **Standard**
This option supports the Kubernetes version for 14 months after the release date. There is no additional cost. When standard support ends, your cluster will be auto upgraded to the next version.

Next

#####

Networking [Info](#)

IP address family and service IP address range cannot be changed after cluster creation.

VPC [Info](#)

Select a VPC to use for your EKS cluster resources. To create a new VPC, go to the [VPC console](#).

vpc-0d51d8961fdd39334 | Default ▼



Subnets [Info](#)

Choose the subnets in your VPC where the control plane may place elastic network interfaces (ENIs) to facilitate communication with your cluster. To create a new subnet, go to the corresponding page in the [VPC console](#).

Select subnets ▼



subnet-05040e2996025489d X
us-west-1a 172.31.16.0/20

subnet-092131509e53d4229 X
us-west-1c 172.31.0.0/20

Clear selected
subnets

Security groups [Info](#)

Choose the security groups to apply to the EKS-managed Elastic Network Interfaces that are created in your control plane subnets. To create a new security group, go to the corresponding page in the [VPC console](#).

Select security groups ▼



DO NOT SELECT DEFAULT SECURITY GROUP

Go to ec2 -> go to security group -> create a security group with all traffic enabled

Security groups [Info](#)

Choose the security groups to apply to the EKS-managed Elastic Network Interfaces that are created in your control plane subnets. To create a new security group, go to the corresponding page in the [VPC console](#).

Select security groups ▼



sg-00ce3c030d07628e1 | myekssg X
eks lab

CLUSTER.CLEAR_S
ELECTED_SECURITY_GROUPS

Choose cluster IP address family [Info](#)

Specify the IP address type for pods and services in your cluster.

☒ IPv4

☐ IPv6

☐ Configure Kubernetes service IP address block [Info](#)

Specify the range from which cluster services will receive IP addresses.

Next

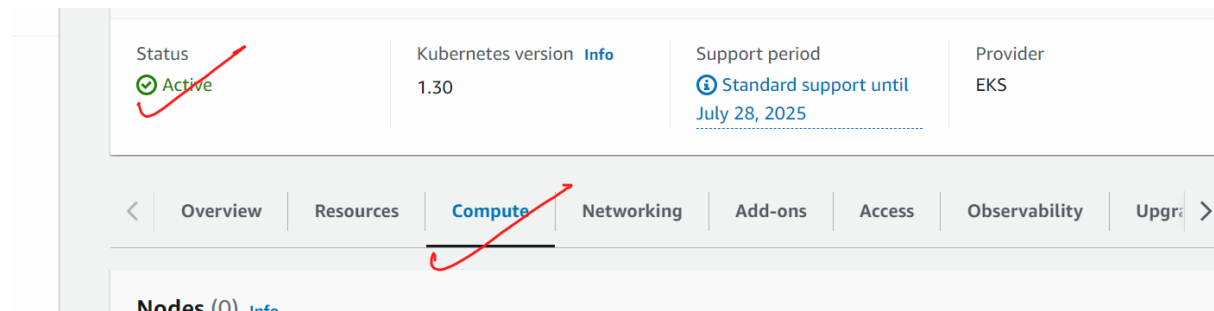
Next

Next

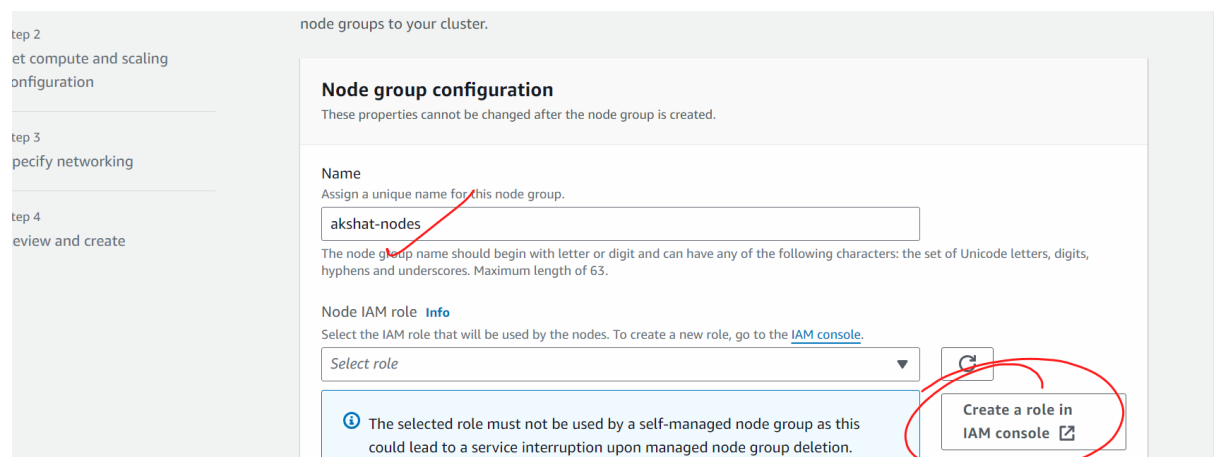
Next

Create

After the EKS cluster is created then we will create the node group



Add node group



(in iam roles -> next -> next (permission will automatically set by aws) -> give some name -> create)

Next

AMI type [Info](#)
Select the EKS-optimized Amazon Machine Image for nodes.

Amazon Linux 2 (AL2_x86_64) ▼

Capacity type
Select the capacity purchase option for this node group.

On-Demand ▼

Instance types [Info](#)
Select instance types you prefer for this node group.

🔍 Enter an instance type

t3.medium ✕
vCPU: 2 vCPUs Memory: 4 GiB Network: Up to 5 Gigabit Max ENI: 3 Max IPs: 18

Disk size
Select the size of the attached EBS volume for each node.

Next

Next

EKS > Clusters > eks-cluster > Node groups > akshat-nodes

akshat-nodes

🔄 Edit Delete

Node group configuration [Info](#)

Kubernetes version 1.30	AMI type Info AL2_x86_64	Status ⋮ Creating
AMI release version Info 1.30.2-20240828	Instance types t3.medium	Disk size 20 GiB

< Details Nodes Health issues 0 Kubernetes labels Update config Kubernetes taints

After the nodes are created we will connect the cluster from the ec2 machine so that we can talk to the cluster

Wait till the time node group is live

My eks service is now working...we need to access the eks from ec2 machine

Launch a ubuntu ec2 machine with all traffic enabled

Connect to the ec2 machine

```
#sudo su
# apt update
#we will now install aws cli
```

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
apt install unzip -y
unzip awscliv2.zip
sudo ./aws/install
```

We will generate access key and secret access keys
In the machine ,

```
# aws configure
Put access key and secret access keys there
```

Method1 to generate access key:

?

⚙️

Asia Pacific (Mumbai) ▼

Leela Gupta (5014-4598-2758) ▲

Leela Gupta

Instance state ▼

Account ID

Account name

Account color

Account

Organization

Service Quotas

Billing and Cost Management

Security credentials

Console Mobile App

Turn on multi-session support

Instance type ▼ | Status change

Instance type	Status change
Instance type	Status change
Instance type	Status change
Instance type	Status change
Instance type	Status change
Instance type	Status change

Click on security credentials

Access keys (0)

Create access key

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

Access key ID	Created on	Access key last used	Region last used	Service last used	Status
No access keys					

As a best practice, avoid using long-term credentials like access keys. Instead, use tools which provide short term credentials. [Learn more](#)

Create access key

Talking:

Create accesskey

Tick the checkbox

keys

Retrieve access key [Info](#)

Access key

If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key

Secret access key

 AKIAJQECPITJJ5QAZWS

 ***** [Show](#)

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

Copy then and put it in notepad

go back to the machine

```
root@ip-172-31-22-171:/home/ubuntu# aws configure
AWS Access Key ID [None]: AKIA6GBMHG2XA33E6OPX
AWS Secret Access Key [None]: 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
Default region name [None]:
Default output format [None]:
root@ip-172-31-22-171:/home/ubuntu#
```

After this is done,,,,,

We will connect this machine with eks cluster

```
aws eks --region example_region update-kubeconfig --name cluster_name
```

(Here replace the example_region with the region where ur eks cluster is created and the cluster name should be the name of the cluster which u have kept)

```
Default output format [None]:
root@ip-172-31-22-171:/home/ubuntu# aws eks --region us-west-1 update-kubeconfig --name eks-cluster
Added new context arn:aws:eks:us-west-1:975050323630:cluster/eks-cluster to /root/.kube/config
root@ip-172-31-22-171:/home/ubuntu#
```

```
sudo apt update
```

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
```

```
chmod +x ./kubectl
```

```

sudo mv ./kubectl /usr/local/bin/kubectl

kubectl version --client

sudo curl --silent --location
"https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(
uname -s)_amd64.tar.gz" | sudo tar xz -C /usr/local/bin

eksctl version

```

```

root@ip-172-31-22-171:/home/ubuntu# kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-172-31-3-130.us-west-1.compute.internal Ready    <none>    61m   v1.30.2-eks-1552ad0
ip-172-31-30-58.us-west-1.compute.internal Ready    <none>    61m   v1.30.2-eks-1552ad0
root@ip-172-31-22-171:/home/ubuntu#

```

```

root@ip-172-31-22-171:/home/ubuntu# kubectl get pods
No resources found in default namespace.

```

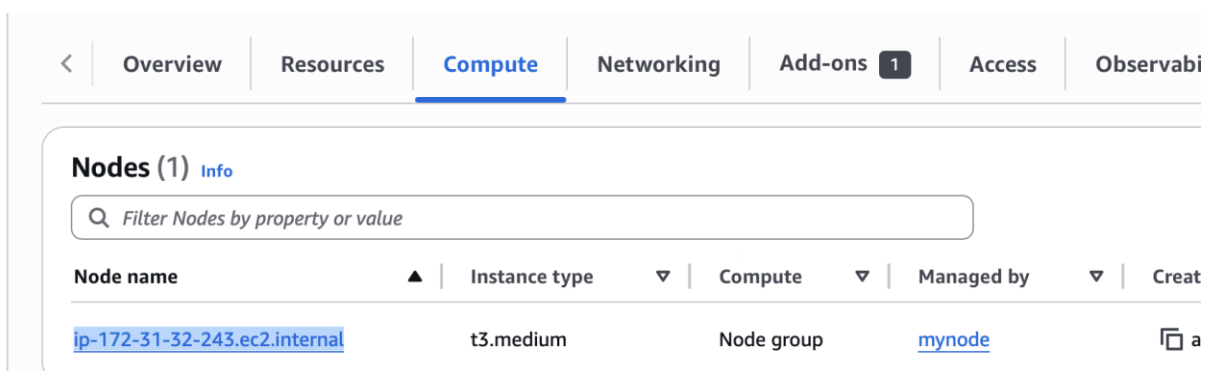
If you get the error 

```

root@ip-172-31-67-175:/home/ubuntu# kubectl get pods
E1115 07:34:27.355951    2326 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get \"https://73EEBBF3944AD5DCBFC28
96.gr7.us-east-1.eks.amazonaws.com/api?timeout=32s\": dial tcp 172.31.38.71:443: i/o timeout"
^C
root@ip-172-31-67-175:/home/ubuntu#

```

Go to eks cluster -> compute -> click on node -> click on instance → security -> click on security group -> edit inbound rule -> delete existing rule and allow all traffic



The screenshot shows the AWS Management Console interface for an EKS cluster. The 'Compute' tab is selected, and the 'Nodes (1)' section is expanded. A table lists the nodes, with one node visible: `ip-172-31-32-243.ec2.internal`, instance type `t3.medium`, and managed by `mynode`. The 'Add-ons' tab shows 1 add-on.

Node name	Instance type	Compute	Managed by	Created
ip-172-31-32-243.ec2.internal	t3.medium	Node group	mynode	

Details

Status ✔ Ready	Kernel version 6.12.53-69.119.amzn2023.x86_64	Created an hour ago
Last transition time an hour ago	Node group mynode	Container runtime containerd://2.1.4
OS (Architecture) linux (amd64)	Instance i-073c8808f5bae3dfc	Kubelet version v1.34.1-eks-c39b1
OS image Amazon Linux 2023.9.20251105	Node auto repair Disabled	Node health issue -

C2 > Instances > i-073c8808f5bae3dfc

Instance details

board

Global View

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ce Types

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js Plans

red Instances

sted Hosts

ity Reservations

ity Manager [New](#)

IMDSv2
Required

Operator
-

Instance ARN
arn:aws:ec2:us-east-1:430118834478:instance/i-073c8808f5bae3dfc

Details

Status and alarms

Monitoring

Security

Networking

Storage

▼ Security details

IAM Role
No roles attached to instance profile: eks-d4cd4320-1d95-a0fa-41af-888b935c33e1

Security groups
sg-0470be878c7ebdb1a (eks-cluster-sg-my-eks-cluster-735157785)

▼ Inbound rules

Owner ID
430118834478

Edit inbound rule

Edit inbound rules

ModifyInboundSecurityGroupRules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID

Type

Protocol

Port range

Source

Description - optional

sgr-0d32a47ba032ce1d3

All traffic

All

All

Custom

0.0.0.0/0

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Save

kubectl get pods

```
root@ip-172-31-67-175:/home/ubuntu# kubectl get pods
No resources found in default namespace.
root@ip-172-31-67-175:/home/ubuntu#
```

```
# vi pod1.yml
```

```
kind: Pod
apiVersion: v1
metadata:
  name: mypod
spec:
  containers:
  - name: c00
    image: ubuntu
    command: ["/bin/bash", "-c", "while true; do echo hello world; sleep 10; done"]
~
~
~
```

```
root@ip-172-31-22-171:/home/ubuntu# kubectl apply -f pod1.yml
pod/mypod created
root@ip-172-31-22-171:/home/ubuntu# kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
mypod     1/1     Running   0           8s
root@ip-172-31-22-171:/home/ubuntu#
```